

JOE SFEIR

Undergraduate - Electrical & Computer Engineering

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EDUCATION

American University of Beirut (AUB)

B.E., Electrical & Computer Engineering

Beirut, Lebanon

Aug 2023 – May 2027 (Expected)

EXPERIENCE

Robotics & Automation Engineering Intern

Omron Corporation

Summer 2026

Dubai, UAE

- Programmed an OMRON i4L SCARA robot using V+ and ACE software, executing complex kinematic path planning and continuous conveyor tracking logic.
- Integrated a Basler GigE camera utilizing AdeptSight machine vision, implementing HSL Color Segmentation and Blob Analysis to dynamically sort moving objects.

Automation Engineering Intern

Eaton Corporation

Winter 2026

Dubai, UAE

- Designed an Automatic Transfer Switch on an Eaton XC-204 PLC using CoDeSys V3.5 & implemented Modbus RTU integration for real-time metering and load-dependents sequencing.
- Programmed logic for pump applications in easySoft, implementing PID thresholds, auto-changeover sequences, and wear leveling algorithms.

PROJECTS

Modeling and Control of a 1-DOF Copter

Fall 2025

- Derived a physics-based non-linear model and refined it using gray-box system identification.
- Designed PID and Lag-Lead controllers using Root Locus and FR methods in MATLAB/Simulink.
- Deployed the final controller on an ESP32, satisfying strict time-domain requirements.

GRAND Decoding Algorithm — RISC-V Assembly

Spring 2025

- Implemented a 5-stage pipelined GRAND soft-decoder in RISC-V Assembly.
- Designed a confidence-vector that reduced clock cycles and improved decoding accuracy.

TECHNICAL SKILLS

Software	AutoCAD, Simulink, CoDeSys, easySoft, CADdy++
Programming	MATLAB, Python, C++, Verilog, RISC-V Assembly
Languages	Arabic (Native), English (Fluent), French (DELFI B2)

ACTIVITIES

Semi-Professional Basketball Player, Tadamon Zouk, *Lebanese Second Division League*

Summer 2025

- Competed in the Lebanese Second Division League while managing academic commitments.

IEEE Student Member, American University of Beirut

Fall 2023 – Present

- Engaged with the engineering community through technical workshops and speaker events.